

Review of the Law of Large Numbers and the Central Limit Theorem

Motivation

- Why can a poll of 1,000 voters predict an election outcome for millions?
- Why do we say a poll has a $\pm 3\%$ margin of error?

1. Law of Large Numbers (LLN)

Definition: If $X_1, X_2, \dots, X_n$ are independent and identically distributed (i.i.d.) random variables with mean μ and finite variance σ^2 , then the sample mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

converges in probability to μ as $n \rightarrow \infty$.

Intuition: As the sample size grows, the average of the sample gets closer to the true population mean. “The more data we collect, the better we approximate the truth.”

Question: What if instead of drawing independent voters, you surveyed the same person 1,000 times? Does the average converge to the truth?

Question: Suppose you survey 10,000 people, all from the same city, as opposed to randomly drawn across the country. What does LLN say?

A **Monte Carlo simulation** is a way to approximate probabilities or expectations by using repeated random draws from a distribution. Instead of solving something analytically (which may be hard), we generate many random samples on the computer, compute the statistic of interest (like the sample mean), and see how it behaves.

2. Central Limit Theorem (CLT)

Definition: Under the same conditions, the normalized sample mean satisfies

$$\sqrt{n} \frac{\bar{X}_n - \mu}{\sigma} \xrightarrow{d} N(0, 1),$$

where σ^2 is the variance of X_i .

Intuition: Even if the original data is not normally distributed, the distribution of the sample mean becomes approximately normal as n increases. This is why normal distributions appear so often in statistics. Averages smooth things out.